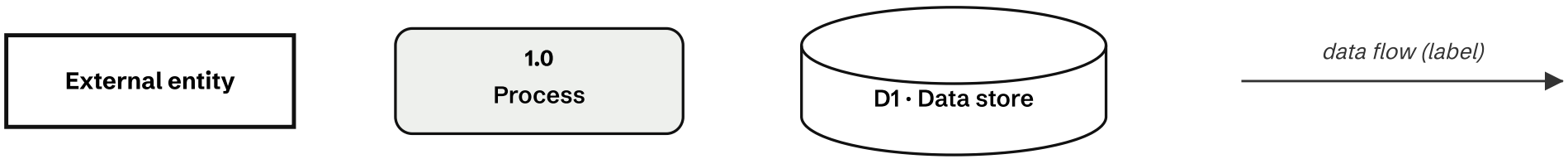
Data Flow Diagrams

How data moves through XOCRISK, the practice-trading website (xocrisk.dpdns.org) and Android app: who sends what, which process handles it, and where it is stored.

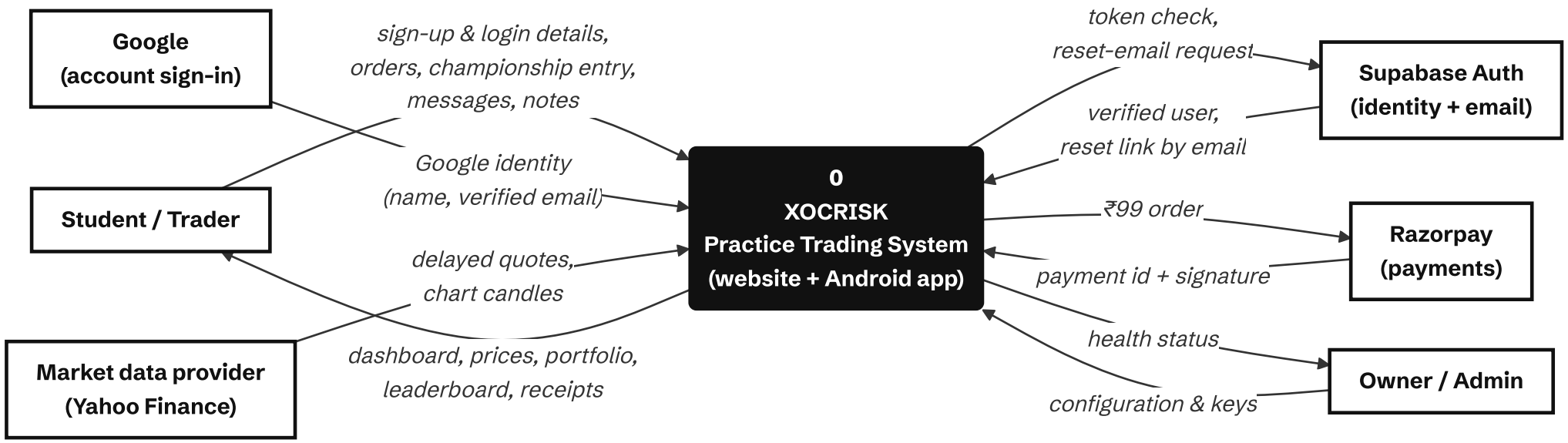
Level 0 · Context diagram 2
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Data dictionary 7

NOTATION



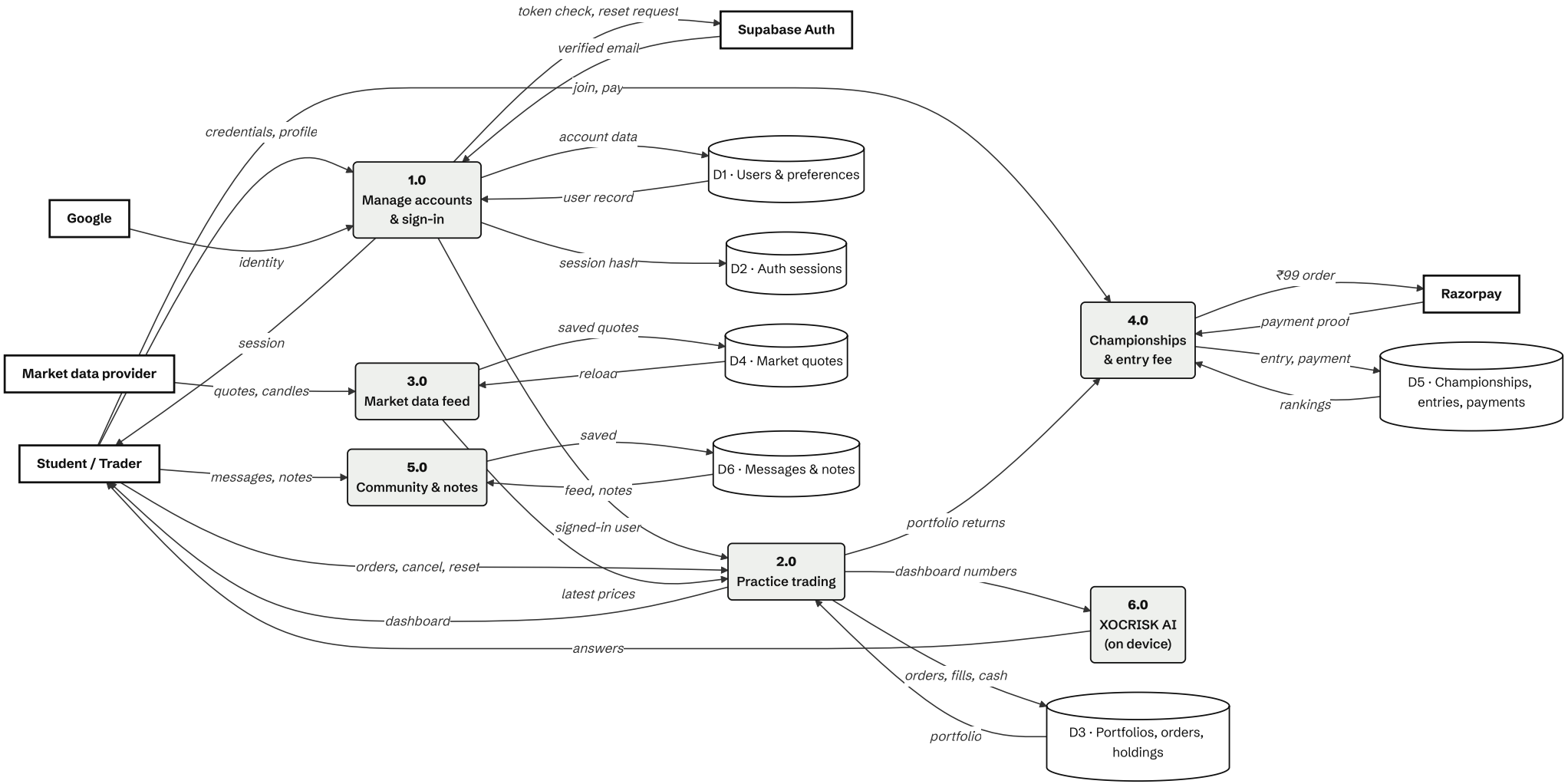
Context diagram

The whole of XOCRISK as one process, with everyone and everything it exchanges data with.



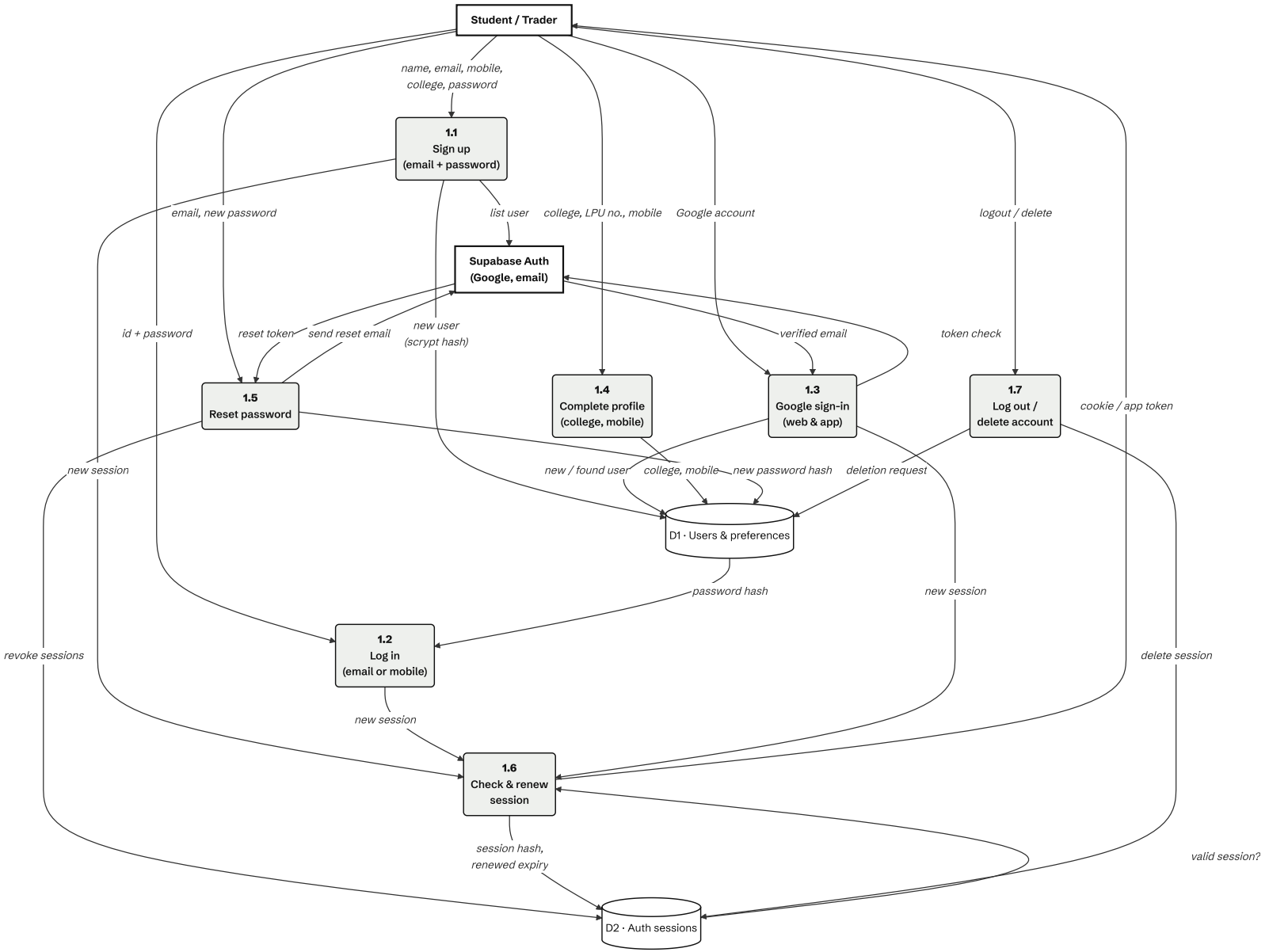
Main processes

The system broken into its six main processes and the data stores they share. D3 groups portfolios, orders and holdings; D5 groups championships, entries and payments.



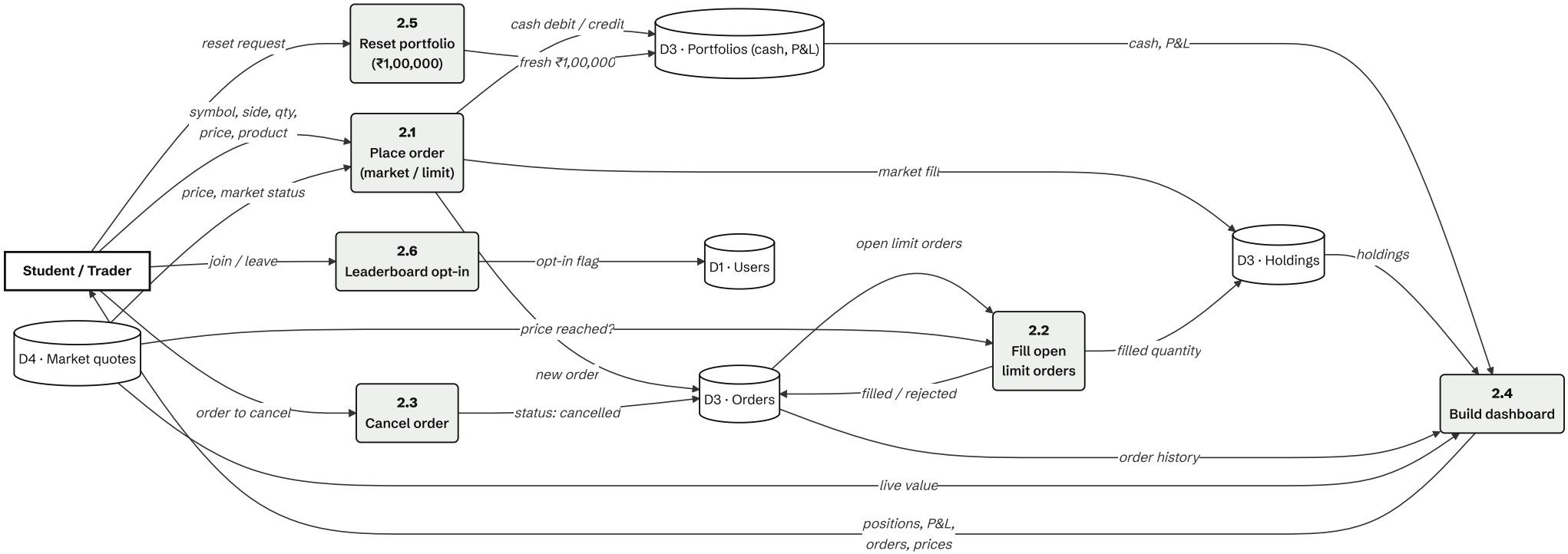
Manage accounts & sign-in

Passwords are stored only as scrypt hashes and sessions only as hashes. Sessions last 180 days and renew while in use; a password reset signs every device out.



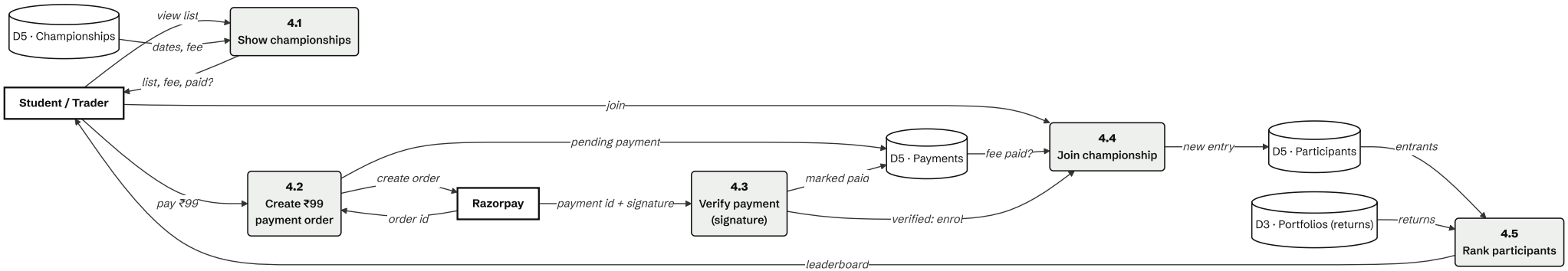
Practice trading

Market orders fill at the current price; limit orders stay open until the price reaches them, or are rejected. Every account starts with ₹1,00,000 of practice money.



Championships & ₹99 entry fee

Entry needs a payment verified on the server with Razorpay's signature, so the fee can't be skipped from the browser.



Data dictionary

Data stores

ID	STORE	DATABASE TABLES (SUPABASE POSTGRES)	MAIN CONTENTS
D1	Users & preferences	`users`	Name, email, mobile, college, LPU no., UID, password hash, privacy & watchlist settings, leaderboard choice
D2	Auth sessions	`auth_sessions`	Hashed session token, user, expiry (180 days, renewed while in use)
D3	Portfolios, orders, holdings	`practice_portfolios, practice_orders, practice_holdings`	Cash, realised P&L, day-start value; orders with side, type, quantity, price, status; holdings per symbol
D4	Market quotes	`market_quotes`	Symbol, price, previous close, high, low, volume, time (delayed, refreshed about every minute)
D5	Championships, entries, payments	`championships, championship_participants, championship_payments`	Dates and ₹99 fee; who entered; Razorpay order and payment ids
D6	Messages & notes	`community_messages, user_notes`	Community chat messages; each user's private notes

External entities

ENTITY	ROLE
Student / Trader	Uses the website or the Android app to practise trading, join championships and chat.
Google	Confirms the person's Google account during Google sign-in.
Supabase Auth	Checks Google sign-in tokens and sends password-reset emails.
Razorpay	Takes the ₹99 championship entry fee and returns a signed payment proof.
Market data provider	Supplies delayed NSE/BSE quotes and chart candles (Yahoo Finance).
Owner / Admin	Sets keys and settings on Vercel and reads the health status.

Processes not expanded

ID	PROCESS	WHAT IT DOES
3.0	Market data feed	Fetches delayed quotes and candles and saves them to D4 about once a minute; every server copy reloads from D4.
5.0	Community & notes	Saves and lists community messages and each user's private notes (D6).
6.0	XOCRISK AI	Runs on the device: prewritten answers filled in with the user's own dashboard numbers. Stores nothing on the server.